



RESEARCH QUESTIONS

Hypothesis 1 — Inhibition

Will a higher concentration of protective lactic acid bacteria inhibit the replication of spoilage microflora, thereby reducing the overall decay rate of the lettuce?

Hypothesis 2 — Kinetics

Will the overall rate of lettuce tissue decay accelerate over time as a direct consequence of the natural mortality and depletion of the protective lactic bacteria?



Bacteria Competition



Decay Kinetics



Shelf-Life Extension

STUDY DESIGN

Formulations

4 distinct bio-coating formulations evaluated across all experimental conditions — including an untreated control group for baseline comparison.

Timeline

21-day post-harvest study with structured data collection at Days 0, 4, and 7 — capturing early, mid, and late-stage decay progression.

Data Acquisition

Multiple dependent variables tracked across all formulations and control groups, including visual tissue integrity, microbial counts, Co2 & O2 emissions, weight loss and structural measurements,

Analytical Framework

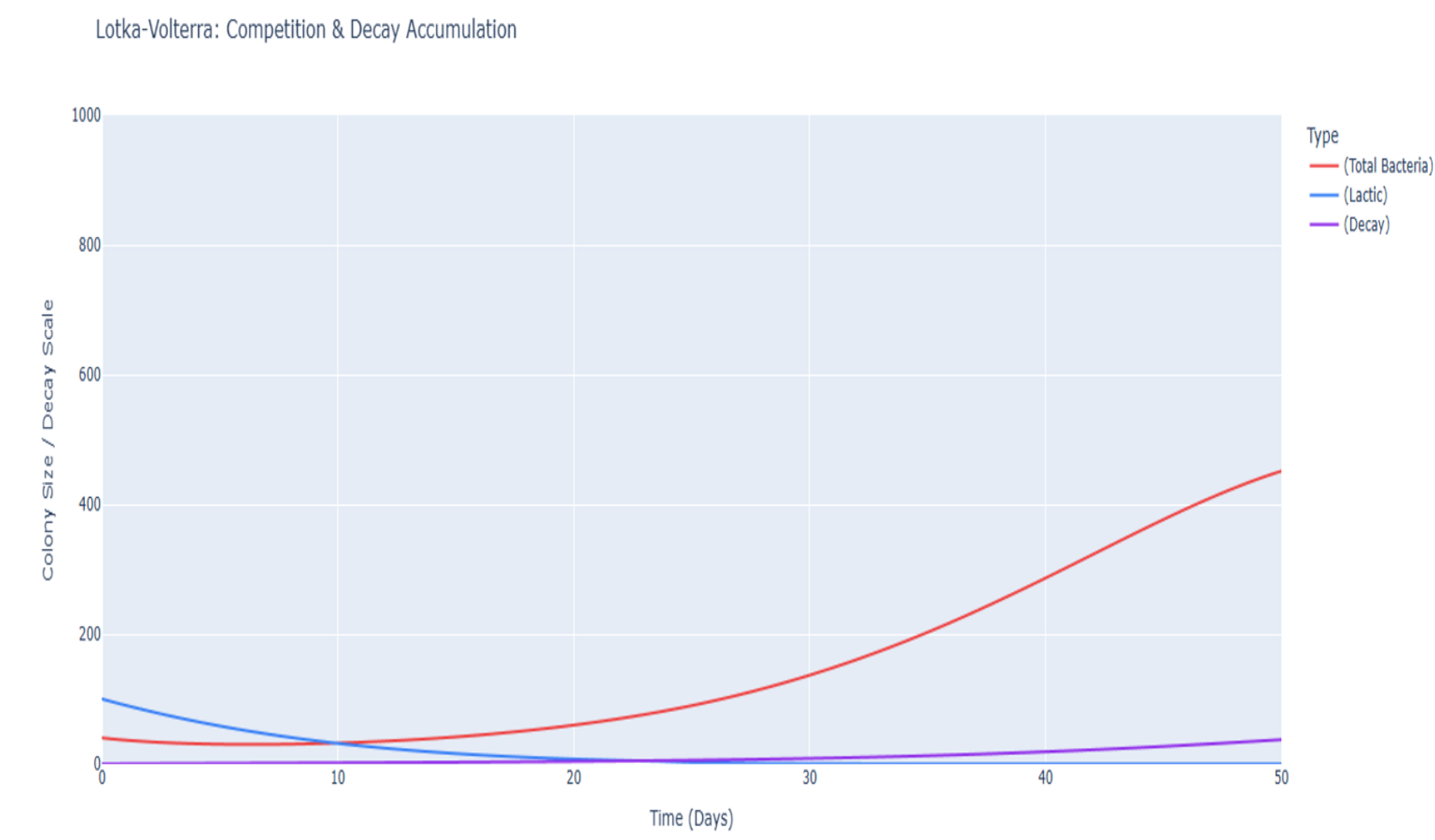
Integration of Principal Component Analysis (PCA), competitive Lotka-Volterra ODE solver, and spatial Agent-Based Modeling (ABM) to synthesize and validate decay kinetics across formulations.

STUDY LIMITATIONS

- Low-density sampling intervals (Days 0, 4, 7 only)
- Evaluation restricted to Romaine Lettuce tissue
- Constant lab conditions — real-world shipping temperature & humidity stress not modeled

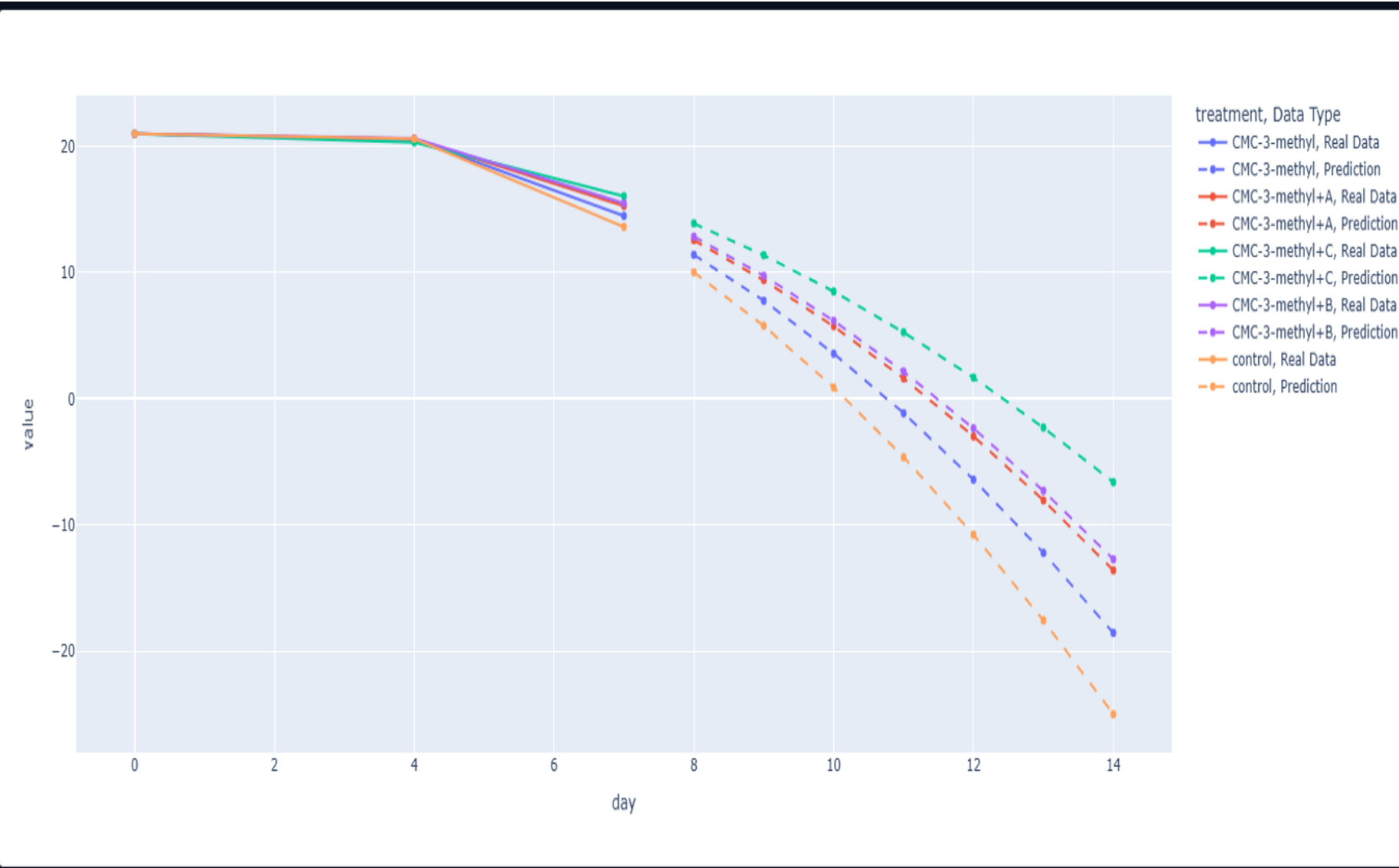
LOTKA-VOLTERRA COMPETITION MODEL

Competitive bacterial dynamics — 50-day simulation of spoilage vs. protective populations



PREDICTIVE SHELF-LIFE ESTIMATION

2nd-degree polynomial regression — decay score by formulation over 14 days



AGENT-BASED MODEL (ABM)

Spatial decay distribution across 50×50 cell grid — microbial competition

Lactic Bacteria

Emitted Ph

Destructive bacteria's

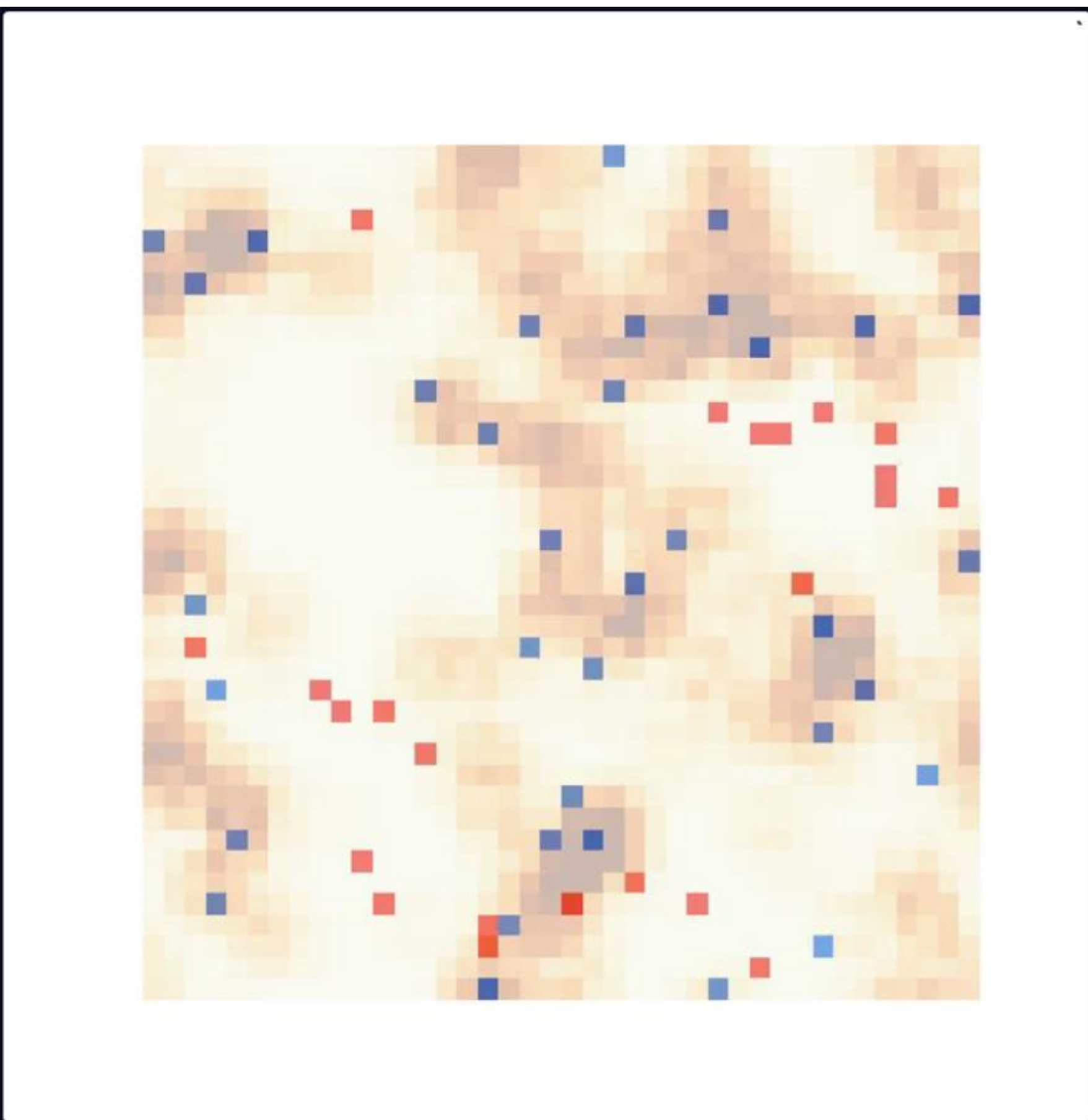
Decayed cells

The spatial ABM distributes 50×50 agents across simulated lettuce tissue. Each agent's state (healthy → decaying → necrotic) is governed by:

- Local lactic acid bacteria density
- Spoilage microflora competition
- Nutrient availability & edge-proximity

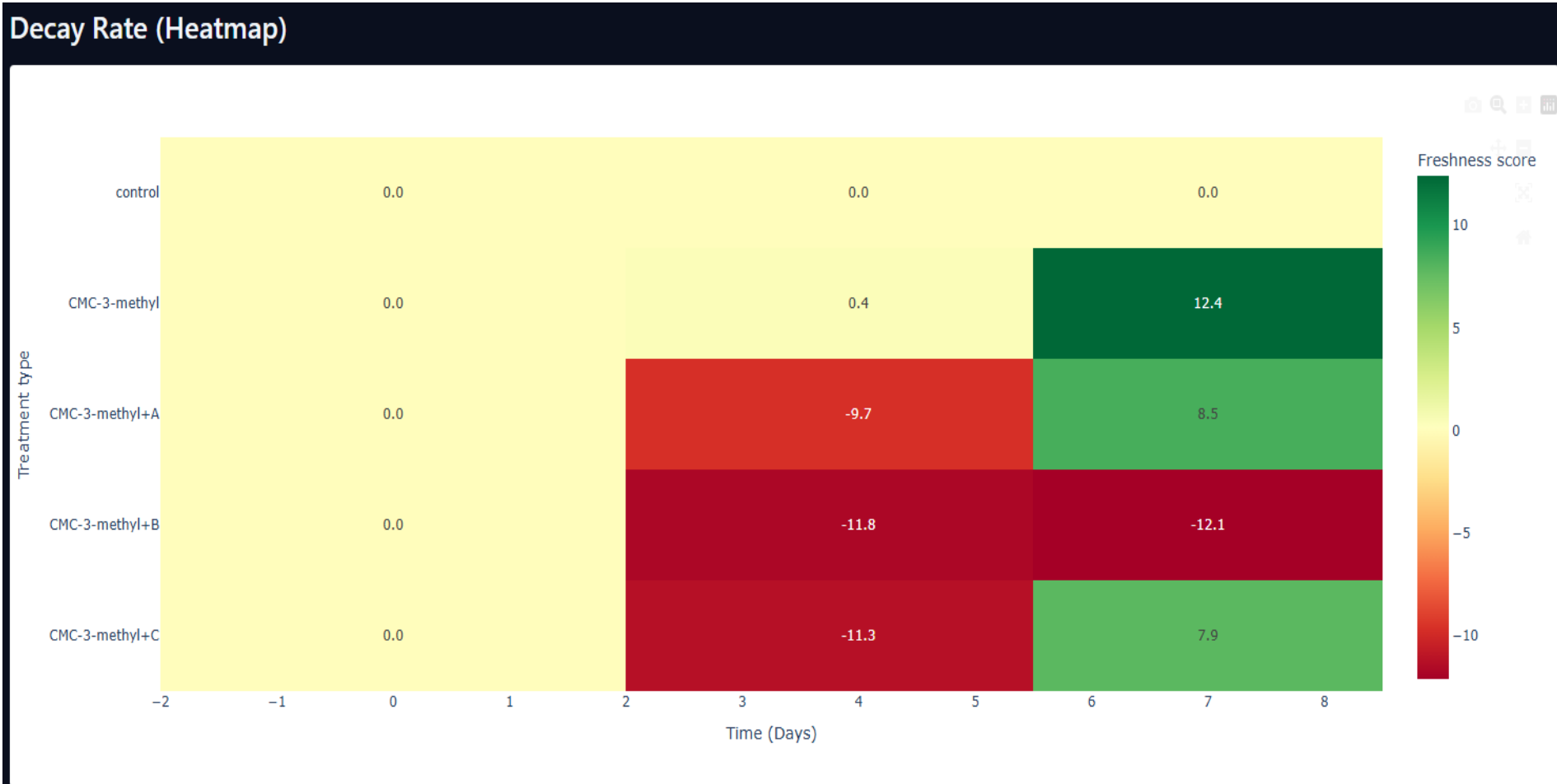
Decay propagates outward from the lettuce edge inward — matching observed real-world spoilage patterns. The model successfully replicates post-harvest tissue decay "tipping points" validated against experimental data.

This approach enables simulation of uneven spoilage across bulk storage containers, giving growers and logistics teams actionable insight into shelf-life variability before shipment.



DECAY RATE HEATMAP (Day 7)

Formulation × Time — decay score comparison



KEY FINDINGS

- CMC-3-methyl+C** exhibits superior decay mitigation, yielding outstanding structural integrity preservation vs. untreated control by Day 7.
- 2nd-degree polynomial regression** delivers highly stable predictive trends on sparse datasets (Days 0, 4, 7) without overfitting — ideal for low-density lab data.
- Lotka-Volterra ODE + ABM** successfully replicate post-harvest tissue decay tipping points, providing accurate qualitative validation for experimental observations.

★ Top Performer: CMC-3-methyl+A

Most effective bio-coating across all measured metrics

CONCLUSIONS & RECOMMENDATIONS



Prioritize Microbial Bio-Coatings

Deploy CMC-3-methyl+A to significantly extend commercial shipping windows and reduce post-harvest losses.



Use Deterministic Fitting Algorithms

Prefer 2nd-degree polynomial regression over complex neural networks for sparse lab datasets.



Embed Real-Time Decay Sliders

Integrate dynamic decay estimation tools in grower-facing logistics software for live shelf-life projections.

